Watcharapon Thodraksa

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SKILLS

Programming Languages: Python, SQL, TypeScript, JavaScript, Java

ML / Computer Vision: PyTorch, OpenCV, semantic segmentation (LR-ASPP, DeepLab), DINOv2, EfficientNet, YOLO (detection, oriented boxes), SAM 2, CLIPSeg, optical flow, audio event detection, scikit-learn, LightGBM

Training / Optimization: knowledge distillation, hard-negative mining, pseudo-labels, ONNX, OpenVINO

Security / On-premises: air-gapped and private-network deployment, VPN-only access, token auth, PII filtering

MLOps/DevOps: Kubernetes (k3s), Kustomize, Helm, Docker, MLflow, Flux GitOps, GitHub Actions, GitLab CI, FastAPI, pytest, Grafana, Prometheus, Kaggle GPU, Railway

LLM / Backend: LangGraph, LangChain, RAG, MCP, LLM evaluation, pgvector, Ollama, Spring Boot, PostgreSQL

EDUCATION

Kasetsart University

Bachelor of Science (B.Sc.) | Major: Computer Science (Co-op Program)

PROFESSIONAL EXPERIENCE

AI/ML Engineer **TRR** 2026 – Present

Computer vision and satellite ML for a sugar mill group in Thailand.

- **CaneSight:** CCTV system that scores every truck dump for dust, leafy cane and burnt cane, and sends the result to the mill's app. Runs on a CPU-only k3s server on the mill's private network.
- **MLOps:** MLflow registry on k3s, gated training on Kaggle, automatic deploy with rollback, Flux GitOps, daily drift monitoring.
- **Burnt-cane classifier:** AUC 0.994 on 483 trucks from days the model never saw (labels from 25,980 weighbridge tickets).
- **CropScan:** harvest detection from Sentinel-1/2 satellite images; F1 0.883 on 8,924 held-out fields.
- **Yield Pro:** rebuilds the crop signal from radar when clouds block the image; 95.7% of values within ±0.10.
- **AI assistant tools (MCP server):** 70 read-only tools that let an LLM answer field questions from live data. Token auth, personal data filtered out (0 leaks in tests), right tool on 30 of 30 questions.
- **CaneGate:** weighbridge truck inspection with Thai plate OCR, shipped as Docker images.

DevOps & Backend Engineer (Intern) **Sycapt** Jan 2026 – May 2026

- Built an on-premises RAG system (LangGraph, BGE-M3, pgvector, Ollama) on Kubernetes for a network with no internet access.
- Ran four LLMs on one 16 GB GPU by swapping models through RAM.
- Ran Kubernetes and Helm deployments, the GitLab CI/CD pipeline with automatic rollback, and Grafana/Prometheus monitoring.

Programmer (Intern) **Sycapt** Nov 2025 – Jan 2026

- Developed backend services for a payment provider handling 1M+ transactions per day (PromptPay, Alipay).
- Built a Kafka connector microservice (Java, Spring Boot) and a map dashboard for 15,423 payment terminals.

PROJECTS

YouTube Shorts Automation:

- Pipeline that reads channel analytics, picks topics, generates and uploads clips on schedule; 3.34M views across 189 clips.

Teaching and Awards:

- Invited lecturer at Kasetsart University: workshops on React Native with the Gemini API and on Git/GitHub.
- 1st Runner-up, KUSE AI Hackathon 2025 · Winner, UI Design-athon 2025.